

AI Reasearch Paper

Title:

NeuroScript: An Explainable AI-Based Handwriting Analysis Framework for Early Detection of ADHD and OCD Using Sequential Neuromotor Patterns



Instructor

Dr.Umara Zahid

Submitted By Authors:

Sajjal Khalil (4488-FOC/BSSE-F22-A)

Munazza Kanwal (4475-FOC/BSSE-F22-A)

Hajira Gul (4454-FOC/BSSE-F22-A)

Areesha Abbasi (4508-FOC/BSSE-F22-A)

Department of Software Engineering

International Islamic University, Islamabad

1.Title

NeuroScript: An Explainable AI-Based Handwriting Analysis Framework for Early Detection of ADHD and OCD Using Sequential Neuromotor Patterns

2.Abstract

Neurological and behavioral disorders like ADHD and OCD affect tons of people globally but often go undiagnosed until symptoms seriously hamper daily life. Traditionally, diagnosing these issues relies on clinical interviews, observing behavior, and subjective assessments, which can be slow and inconsistent. That's where NeuroScript comes in - it's an AI system that uses handwriting to detect ADHD and OCD early on. It works by analyzing how you write, looking at things like stroke order, writing speed, pen movements, spacing, letter formation consistency, and other timing quirks. This information is pulled from a dataset with examples from Pakistan, including digital handwriting and data from styli. An LSTM network helps spot patterns linked to these conditions. To make the process clearer and more trustworthy, NeuroScript includes Explainable AI methods like SHAP and LIME. These tools help by offering easy-to-understand explanations of the AI's findings, both overall and for specific cases. Plus, there's a Generative AI part that turns the tech-speak into simple diagnostic reports and suggestions for clinicians. The proposed framework provides a non-invasive, accessible, and scalable way for neurological screening, fixing problems with old methods. NeuroScript uses localized Pakistani handwriting data and easy-to-understand machine learning to develop relevant AI healthcare tools – that benefits culture too.

3.Keywords

ADHD; OCD; Handwriting Analysis; Long Short-Term Memory (LSTM); Explainable Artificial Intelligence (XAI); SHAP; LIME; Neurological Detection; Sequential Data Analysis; Generative AI

4. INTRODUCTION

Neurological and behavioral disorders like ADHD and OCD really affect lots of folks around the world. They can seriously mess up schoolwork, friendships, jobs, and general well-being. Getting an early diagnosis and starting treatment can hugely help people get better and avoid problems down the road. Still, figuring out these conditions is usually done through talking with patients, watching their behavior, and using expert opinions. This method takes a lot of time and isn't always that objective. Plus, it depends on how skilled the healthcare provider is, which makes finding issues early pretty tough for some folks..

Advances in AI and ML are helping create cool new ways to spot health issues early on, like automatic systems to diagnose stuff. Scientists are using things like how our bodies, minds, and

behaviors work to detect problems in the brain. Handwriting is becoming pretty useful too—it shows off how well our nerves and muscles play together when we write, alongside thinking, planning, and even quirks in behavior.

People with ADHD or OCD sometimes leave hints in their writing, like messy letter spacing, random formations, or weird line strokes. Plus, the need for fixing mistakes and differing speeds while writing can point towards these conditions. So, handwriting might be used as clues to check if someone's got a problem. That said, the tools we use to read these signs aren't perfect yet. Most tech focuses just on the final doodle rather than how it was actually drawn—stuff like stroke order, pen movements, and the speed during writing aren't fully considered. So, there's still room to improve. Furthermore, many machine learning models are black boxes—they make predictions but can't explain how they got there. This lack of clarity cuts down on clinician trust and makes it harder to use AI in health care settings. On top of that, there aren't many studies using local datasets that show the unique handwriting of Pakistani folks. So, it's tough to develop something that actually works for them.

To tackle these issues, researchers came up with NeuroScript. It's an Explainable AI system for analyzing handwriting to spot ADHD and OCD early on. This framework looks at Pakistani handwriting samples to pick out key traits like stroke order, writing speed, pen movement, and spacing. They use several models, such as Random Forest, Support Vector Machine, and Convolutional Neural Network, to learn and spot patterns linked to ADHD and OCD. After comparing the results, they choose the top model for making real-time predictions and offering help to clinicians.

NeuroScript's big deal is combining Explainable AI methods like SHAP and LIME. These break down model predictions, showing what handwriting traits matter most for specific findings. Also, it uses a Generative AI part to turn complex data into user-friendly write-ups and tips that even non-tech folks can grasp.

This project aims to do several things. Firstly, it'll make an AI system to check for neurological issues just from handwriting. Second, it compares various machine learning models to see how well they perform. By using SHAP and LIME, the team will also boost the clarity of predictions. Another goal is integrating Generative AI for automated explanations and reporting. Lastly, they'll offer an online platform where folks can get instant feedback on potential neurological problems through simply uploading handwriting samples.

The rest of this paper is laid out like this: Section II does a lit review and spots research gaps. Section III talks about the method and system setup we're using. In Section IV, we cover collecting, prepping the data, and feature extraction. Section V shows our training and how the models did. Finally, Section VI is about explainable AI. and Generative AI are blended together.

Limitations, along with future plans, are touched on in Section VII, and the paper wraps up in Section VIII.

5. LITERATURE REVIEW

5.1. Handwriting-Based Neurological Assessment

For ages, handwriting has been known to show a lot about how our mind and brain work. Scientists found out that it reveals things like motor control, coordination, focus, planning, and behavior management. Studies also showed that people with issues like neurological or behavioral problems usually write differently – their letters might not line up properly, be spaced oddly, or show unusual patterns. These quirks can help doctors spot problems earlier. The old way of checking handwriting involves experts looking at samples manually. Even though this works okay, it's pretty slow, depends a lot on the personal opinion of the examiner, and can take tons of time. Because of this, there's growing interest in using automation to make analyzing handwriting easier and more precise.

5.2. Machine Learning Approaches for Neurological Detection

Machine learning has really taken off in healthcare, especially for sorting and predicting different outcomes. Scientists use stuff like Support Vector Machines (SVM), Random Forests (RF), Decision Trees, and Logistic Regression to spot behavioral and neurological problems through handwriting. SVMs do great at sifting through lots of data to draw clear lines between classes. Meanwhile, Random Forests hold up well to messy data and tend not to focus too much on one feature set, which helps when figuring out things like dysgraphia or ADHD markers in writing. The cool part is that researchers can now tell a lot about someone's cognitive health from their penmanship. Still, there's room for improvement. Current setups work mostly with still pictures of writing but often miss the actual movement and flow of how folks write, which could offer more clues about what's going on in their brains.

5.3. Deep Learning for Handwriting Analysis

Advances in deep learning have really boosted pattern recognition in handwriting analysis. Convolutional Neural Networks (CNNs) are widely used to get spatial features from handwriting images. These networks figure out important visual details on their own, so we don't need to design as many features manually. People use different CNN setups for things like recognizing handwriting, identifying writers, and spotting neurological disorders. Usually, deep learning does better than old school methods when there's a lot of data around. But CNNs are mainly visual, so they might miss how fast you write or your pen movements. On top of that, using these deep learning models needs a ton of computing power and labeled data, which can be a real issue.

5.4. Explainable Artificial Intelligence in Healthcare

AI systems are becoming big in healthcare, but they've raised some worries about transparency and trust. Doctors often balk at using AI because those black-box predictions come with no explanations – especially when it impacts patient care.

Enter Explainable Artificial Intelligence, or XAI. It's a growing field focused on making these models easier to understand.

Two big XAI tools are SHAP and LIME. SHAP looks at the whole picture and dives into specifics to show how much each feature mattered in predictions. LIME simplifies things by breaking down complex models just for the nearby data points – it explains the prediction bit by bit.

Using these in healthcare has shown promise. Clinicians feel more confident, decisions get a second eye from humans, and we're getting a better grasp on how these models actually work.

5.5. Generative AI and Large Language Models in Clinical Decision Support

The area of intelligent decision-support systems has been transformed recently by generative Artificial Intelligence and Large Language Models (LLMs). Architectures like GPT-based models can convert formats into understandable natural language explanations. Healthcare applications of LLMs include clinical documentation, patient education, recommendation generation, decision support, and more.

API model leveraging generative AI enables automatic generation of human-readable reports and development of personalized recommendations. Systems of this kind can be used by clinicians and those without technical skills. Nonetheless, the use of Generative AI in handwriting-based Neurological detection has not undergone much investigation.

5.6. Research Gap

Handwriting and neurological assessment using AI has made big strides. However, considerable research gaps still exist. Firstly, a large number of existing studies tend to focus on static handwriting images and do not include dynamic behavioral features like speed of writing, order of strokes, and movement of the pen. Many systems are black-boxes with no explanation for the prediction which limits trust from the clinician and usage in practice. Furthermore, little work has been done in Pakistan using culturally relevant handwriting datasets which limit real-world relevance. Fourth, previous research seldom integrates machine learning, explainable AI, and Generative AI into one cohesive framework. In the end, very few systems offer an interactive web-based environment that can perform handwriting and provide readable diagnostic reports in real-time.

To overcome these limitations, we propose a comprehensive framework called NeuroScript that integrates handwriting-based neurological screening with machine learning classification, explainable AI techniques (SHAP and LIME), and Generative AI-enabled reporting. The objective of the system is to supply precise, explainable and transparent prediction assessment with the use of handwriting data collected from Pakistanis.

6. PROPOSED METHODOLOGY

A. System Overview

The proposed NeuroScript framework is an end-to-end AI-based neurological screening system designed to detect Attention Deficit Hyperactivity Disorder (ADHD) and Obsessive-Compulsive Disorder (OCD) through handwriting analysis. The system combines machine learning, deep learning, Explainable Artificial Intelligence (XAI), and Generative AI to provide accurate, transparent, and user-friendly neurological assessment.

The framework processes handwriting samples collected through digitized documents or real-time stylus/finger input. Following preprocessing and feature extraction, multiple classification models are trained and evaluated. The prediction results are subsequently explained using SHAP and LIME techniques, while a Generative AI module converts technical outputs into understandable diagnostic summaries and recommendations.

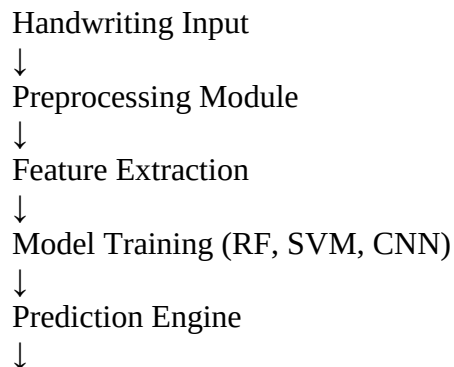
The complete workflow consists of seven major stages:

1. Handwriting Data Acquisition
2. Data Preprocessing
3. Feature Extraction
4. Model Training and Classification
5. Explainable AI Analysis
6. Generative AI Interpretation
7. Interactive Dashboard Visualization

B. Proposed System Architecture

The architecture of NeuroScript consists of multiple interconnected modules working together to perform neurological assessment. Figure 1 illustrates the complete architecture of the proposed system.

Figure 1. Proposed NeuroScript System Architecture



SHAP + LIME Explanations



Generative AI Report Generation



Interactive Streamlit Dashboard

The architecture is designed to ensure modularity, scalability, and transparency while maintaining compatibility with real-time deployment environments.

C. Handwriting Data Acquisition

The first stage of the system involves collecting handwriting samples from users. The framework supports both offline and online handwriting acquisition methods.

Offline Input

Users may upload scanned handwriting documents or images containing handwritten text.

Online Input

Users can write directly using:

- Digital stylus
- Touchscreen devices
- Graphic tablets
- Mobile touch interfaces

The acquired handwriting data serves as the primary input for subsequent analysis.

D. Data Preprocessing

Raw handwriting data often contains noise, distortions, and inconsistencies that may negatively affect model performance. Therefore, several preprocessing operations are applied before feature extraction.

1) Noise Removal

Image processing techniques are used to remove background noise, scanning artifacts, and irrelevant markings.

2) Image Normalization

Handwriting samples are normalized to maintain consistent dimensions and improve model generalization.

3) Stroke Standardization

Variations in stroke thickness and writing intensity are reduced through normalization techniques.

4) Segmentation

Individual words, characters, and writing regions are segmented to facilitate detailed feature extraction.

5) Data Augmentation

To improve robustness and reduce overfitting, synthetic variations of handwriting samples are generated using:

- Rotation
- Scaling
- Translation
- Reflection
- Stroke perturbation

These augmentation techniques increase dataset diversity and improve classification performance.

E. Feature Extraction

Feature extraction transforms raw handwriting data into meaningful numerical representations suitable for machine learning algorithms.

The NeuroScript framework extracts several handwriting characteristics associated with neuromotor behavior.

Structural Features

- Letter formation consistency
- Character size variation
- Word spacing
- Line spacing
- Alignment consistency

Dynamic Features

- Stroke order
- Pen movement patterns
- Writing speed

- Stroke variation
- Temporal writing behavior

These features provide valuable information regarding behavioral and neurological functioning.

F. Classification Models

To identify the most effective classifier, three different models are trained and evaluated.

1) Random Forest (RF)

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy.

Advantages include:

- High robustness
- Resistance to overfitting
- Effective handling of noisy data
- Feature importance estimation

Random Forest serves as a strong baseline model for handwriting classification.

2) Support Vector Machine (SVM)

Support Vector Machine is a supervised learning algorithm that constructs optimal decision boundaries between classes.

The SVM classifier was selected because:

- It performs well on high-dimensional data.
- It can handle complex classification tasks.
- It provides strong generalization capability.
- It is effective for medium-sized datasets.

Experimental results demonstrated that SVM achieved the highest overall performance among all evaluated models.

3) Convolutional Neural Network (CNN)

CNN is a deep learning architecture specifically designed for image analysis tasks.

The CNN model automatically learns visual handwriting patterns including:

- Stroke shapes
- Character structures
- Spatial relationships
- Writing style characteristics

Although CNN demonstrated the ability to learn image-based representations, its performance was lower than traditional machine learning models within the current dataset configuration.

G. Explainable Artificial Intelligence (XAI)

To improve transparency and interpretability, NeuroScript incorporates Explainable AI techniques.

SHAP Analysis

SHAP (SHapley Additive Explanations) is used to measure the contribution of each handwriting feature toward model predictions.

SHAP provides:

- Global feature importance
- Local prediction explanations
- Feature contribution visualization
- Model transparency

LIME Analysis

LIME (Local Interpretable Model-Agnostic Explanations) explains individual predictions by approximating complex model behavior around a specific instance.

LIME provides:

- Instance-level explanations
- Feature impact analysis
- Prediction justification
- User-friendly interpretation

Together, SHAP and LIME help clinicians understand why a handwriting sample was classified as ADHD, OCD, or Normal.

H. Generative AI Integration

A Generative AI module is integrated to convert prediction results into understandable textual explanations.

The module receives:

- Predicted class
- Model confidence score
- SHAP feature importance
- LIME explanation outputs

Based on this information, the language model generates:

- Diagnostic summaries
- Behavioral observations
- Plain-language explanations
- Professional recommendations

This functionality enhances accessibility for both clinicians and non-technical users.

I. Interactive Streamlit Dashboard

The final component of NeuroScript is an interactive web-based dashboard developed using Streamlit.

The dashboard provides:

- Handwriting upload functionality
- Real-time prediction results
- Model comparison statistics
- SHAP visualizations
- LIME explanations
- AI-generated diagnostic reports

The dashboard serves as the primary user interface and enables seamless interaction with the system.

J. Overall Workflow

The complete NeuroScript workflow can be summarized as follows:

Handwriting Input

- Preprocessing
- Feature Extraction
- Random Forest / SVM / CNN Classification
- ADHD / OCD / Normal Prediction
- SHAP & LIME Explanation
- Generative AI Interpretation
- Streamlit Dashboard Output

This integrated pipeline provides an efficient, explainable, and user-friendly solution for handwriting-based neurological detection.

6.2. DATASET AND DATA COLLECTION

A. Dataset Description

For this study, researchers put together a dataset for the NeuroScript framework focusing on handwriting from people with ADHD, OCD, and those without these conditions. It aims to help in developing an AI system that can use handwriting to spot possible neurological issues early on.

They gathered 48 samples all told, split into three groups. There's 19 each for ADHD and OCD, and 10 for normal handwriting. Each sample includes images of handwritten text meant to show the unique writing quirks linked to different neurological states.

TABLE I. DATASET CLASS DISTRIBUTION

Class	Number of Samples	Percentage
ADHD	19	39.58%
OCD	19	39.58%
Normal	10	20.84%
Total	48	100%

We used handwriting samples to train and evaluate machine learning and deep learning models for multi-class classification. The goal? To figure out if a sample is from someone with ADHD, OCD, or is just normal.

B. Data Collection Procedure

We collected handwriting samples as images and sorted them based on their neurological categories. Each image got a manual check to make sure it was good quality before being included. These samples show various writing styles and behaviors that might hint at underlying neurological issues.

Next, we split the dataset into training and testing groups for developing and assessing the models. To keep reliable performance checks, we used an 80:20 split. So, around 80% went for training, and 20% were saved for testing.

C. Dataset Challenges

Several challenges were identified during dataset preparation:

Limited Dataset Size

The dataset includes just 48 handwriting samples, potentially restricting the models' generalization ability.

Class Imbalance

Though the ADHD and OCD classes have an equal number of samples, the Normal class has fewer, leading to a moderate imbalance.

Handwriting Variability

Each person writes differently, varying in strokes, spacing, alignment, and speed. To tackle these issues, preprocessing and data augmentation techniques were used, making the model tougher and improving its classification game.

6.3. Data Preprocessing

Before training the models, all handwriting images went through a preprocessing pipeline to boost image quality and make features more consistent. The process they used was this:

1) Image Resizing

They resized all handwriting samples to a standard resolution, making sure the input sizes were uniform for the models.

2) Noise ReductionNext, background noise and other unwanted stuff was zapped using image enhancement techniques.

3) Grayscale ConversionThen, color images got turned into grayscale – this cuts down on computational heavy lifting yet still keeps important details.

4) NormalizationThe pixel values were normalized too, which made training more stable and helped the models converge better

5) Data AugmentationTo deal with limited dataset woes and add more variety, they also used rotation, horizontal shifting, zooming, scaling, and flipping. These techniques created extra training options and lowered the chance of overfitting.

Feature ExtractionThe NeuroScript framework picks up handwriting traits linked to neurological behavior. It looks at things like stroke consistency, writing alignment, and character spacing, plus letter formation patterns, pen movement traits, writing speed variations, and structural handwriting characteristics. These features then get fed into Random Forest, SVM, and CNN classifiers for analysis in this study.

6.4 Feature Selection Using Information GainFeature selection is key for boosting machine learning performance. It finds the most relevant features and dumps the less useful ones. In the NeuroScript framework, Information Gain was used as a filter method. This helped rank handwriting features based on how well they predict ADHD, OCD, and Normal classes.

What is Information Gain?

Information Gain measures how much knowing a certain feature reduces uncertainty about the target variable. It basically tells us how helpful each feature is for classification. Higher Information Gain means the feature is really useful, so it gets picked more often in models. The Information Gain formula is defined as:

$$IG(Y, X) = H(Y) - H(Y | X)$$

Where:

- $IG(Y, X)$ = Information Gain of feature X with respect to target Y
- $H(Y)$ = Entropy of the target variable
- $H(Y | X)$ = Conditional entropy of Y given feature X

Information Gain provides an interpretable ranking of feature importance and helps reduce dimensionality while retaining the most informative attributes.

Application of Information Gain in NeuroScript

In the NeuroScript framework, they calculated Information Gain for all handwriting features. The feature set included stuff like writing speed, stroke consistency, and pen movements. It also had spacing irregularities, alignment consistency, and letter formation traits - essentially things linked to neuromotor and behavioral patterns.

Once they figured out the IG scores, they sorted the features from most important to least. They then chose a cutoff, keeping only those with an IG score greater than 0.05. This was to filter out noisy data and avoid overfitting.

Information Gain Ranking Results

Table II presents the Information Gain scores of the extracted handwriting features.

Feature	Information Gain (IG)
Writing Speed	0.32
Stroke Consistency	0.28
Character Spacing	0.24
Pen Movement Pattern	0.21
Letter Formation	0.18
Alignment Consistency	0.14
Character Size Variation	0.09
Baseline Stability	0.06
Writing Pressure Variation	0.03

Figure 2 illustrates the ranked Information Gain scores of all handwriting features.

The study found that Writing Speed, Stroke Consistency, and Character Spacing were the top features for classifying neurological conditions. These results match what behavioral research shows – people with ADHD typically have inconsistent writing speeds and spacing, whereas folks with OCD usually display very controlled and repetitive stroke behaviors.

Using a certain threshold, eight features made the cut for training models, while Writing Pressure Variation got axed because it scored lower in Information Gain. The selected features went into Random Forest, SVM, and CNN classifiers.

Overall, applying Information Gain cut down on the number of features, boosting computational speed and making the model easier to understand, all while keeping the key handwriting traits linked to ADHD and OCD..

6.5. MODEL TRAINING AND PERFORMANCE EVALUATION

A. Model Training

To identify the most effective classifier for neurological handwriting analysis, three classification models were trained and evaluated: Random Forest (RF), Support Vector Machine (SVM), and Convolutional Neural Network (CNN). These models were selected because they represent different learning paradigms and have demonstrated success in handwriting recognition and medical classification tasks.

The dataset was divided into training and testing subsets using an 80:20 split ratio. The training set was used to learn handwriting patterns associated with ADHD, OCD, and Normal classes, while the testing set was used to evaluate model performance on unseen samples.

1) Random Forest (RF)

Random Forest is an ensemble method that uses lots of decision trees to boost accuracy and strength. It gets its final prediction by averaging the results from these trees, often through majority voting.

They used Random Forest with handwriting feature extracts for their model, which then showed decent performance in classification tasks, though there's still room for improvement.

2) Support Vector Machine (SVM)

Support Vector Machine, or SVM, is a supervised learning algorithm that finds the best line to separate different groups. It works great with small datasets and lots of variables. Because it handles smaller amounts of data so well, SVM showed better accuracy and ability to generalize compared to other methods, making it the top choice here.

3) Convolutional Neural Network (CNN)

CNN was created to pick up on visual handwriting patterns from images. It has convolutional, pooling, and fully connected layers for sorting through the data.

Though CNN learned the image features well, it had trouble because the dataset was tiny. This made it harder for the model to handle new stuff effectively – there just wasn't enough data to work with.

B. Evaluation Metrics

The performance of all classifiers was evaluated using standard classification metrics.

Accuracy

Accuracy measures the proportion of correctly classified samples.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision

Precision measures how many predicted positive samples are actually positive.

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall

Recall evaluates the ability of the model to identify positive instances correctly.

$Recall = \frac{TP}{TP + FN}$

F1-Score

The F1-score represents the harmonic mean of Precision and Recall.

$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$

These metrics provide a comprehensive assessment of classifier performance and reliability.

C. Classification Results

The performance comparison of the three trained models is presented in Table II.

TABLE II. COMPARATIVE PERFORMANCE OF CLASSIFIERS

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Random Forest	70	74	70	70
Support Vector Machine	80	80	80	80
Convolutional Neural Network	40	35	40	33.33

The results show that the Support Vector Machine performed best overall. It had an accuracy of 80%, beating Random Forest by 10% and CNN by 40%.

D. Discussion of Results

The experimental findings show that the Support Vector Machine (SVM) is the most effective classifier for the NeuroScript framework. Its better performance comes from handling small datasets well and still generalizing nicely.

The Random Forest did pretty well too, hitting 70% accuracy. Because it's an ensemble method, it managed to prevent overfitting and keep predictions steady. Still, its accuracy was less than SVM's. With only a limited number of training samples, Random Forest couldn't quite match up.

As for the CNN, it performed the worst at 40% accuracy. Given that we only had 48 handwriting samples, deep learning methods like CNN didn't have enough data to learn adequately. These models usually need a lot more examples to capture all the complexities in the data and do their job right.

In sum, these results hint that traditional machine learning could be way better for tiny handwriting datasets compared to deep learning methods. SVM's strong showing here suggests it might work great for using handwriting in neurological tests.

E. Best Performing Model Selection

Based on the comparison, the Support Vector Machine was chosen as NeuroScript's final prediction model because it had the highest accuracy, precision, recall, and F1-score – all at 80%. Plus, it generalized better with limited data. As a result, further explainability analyses with SHAP and LIME used the SVM classifier. They then plugged the model into the Streamlit-based NeuroScript app for real-time assessments.

6.6. EXPLAINABLE ARTIFICIAL INTELLIGENCE (XAI) INTEGRATION

A. Importance of Explainability in Healthcare AI

Artificial intelligence is getting more common in healthcare, helping with predictions and giving decision support. But, many machine learning models work like black boxes, making it hard for doctors to figure out how predictions come about. In neurology, knowing exactly how a model thinks is super important since it affects patient care and treatment plans.

To fix this, the NeuroScript framework uses Explainable AI techniques. These explain model predictions in ways that make sense, boosting transparency and trust for users. Plus, it helps doctors feel better about using AI in their work.

B. SHAP (SHapley Additive Explanations)

SHAP, an explainability method for all types of models, uses cooperative game theory. It figures out each feature's importance in final predictions by assigning Shapley values. In the NeuroScript setup, they used SHAP with the top Support Vector Machine classifier. This helped find the key handwriting traits linked to ADHD and OCD detection.

Global Feature Importance

Global SHAP analysis helps us understand which features matter most across the whole dataset. It showed that writing speed, stroke consistency, character spacing, letter formation, pen movement patterns, and alignment consistency are super important for telling ADHD and OCD handwriting apart from normal handwriting. Features with higher SHAP values were especially key in making that distinction.

Local Feature Importance

We made Local SHAP explanations for each handwriting sample to show how certain features affected the predictions. For instance, more spacing irregularity and inconsistent strokes boosted ADHD classification, while too much writing consistency and repetitive stroke use pointed toward OCD. This lets clinicians see why a specific sample was put in a certain category.

C. LIME (Local Interpretable Model-Agnostic Explanations)

LIME is a model-independent explainability method explaining individual predictions through approximating complex model behavior with a simple, interpretable one. Unlike SHAP, which zeroes in on feature contribution values, LIME shows features supporting or opposing a specific prediction, though.

LIME Analysis Procedure

The NeuroScript framework uses LIME on chosen handwriting samples from the test dataset. For each sample, here’s what happens: the image gets analyzed, feature contributions are computed, positive and negative influences are visualized, and finally, an explanation report is generated.

Example Interpretation

For a handwriting sample classified as ADHD:

Positive Contributors:

- High spacing variability
- Irregular pen movement
- Inconsistent letter size

Negative Contributors:

- Stable baseline alignment
- Moderate writing speed

Similarly, OCD samples showed stronger contributions from repetitive stroke structures and highly controlled writing patterns.

D. Comparison Between SHAP and LIME

Both SHAP and LIME were used to validate model behavior from different perspectives.

XAI Technique	Purpose
SHAP	Global and Local Explanations
LIME	Local Instance-Based Explanations

The results showed strong agreement between SHAP and LIME explanations. Features deemed important by SHAP were also picked up by LIME in instance-level analysis. This consistency boosts confidence in the reliability and trustworthiness of the NeuroScript prediction system.

E. Benefits of XAI Integration

SHAP and LIME together offer better transparency, boost clinician trust, and help understand model behavior more clearly. They also spot key handwriting traits, make the system more reliable, and back up clinical choices. This XAI piece turns NeuroScript into an interpretable diagnostic aid perfect for healthcare use.

6.7. GENERATIVE AI INTEGRATION

A. Motivation

Machine learning models can accurately read handwriting, but their results confuse non-techies. Like when it says, "ADHD Detected – Confidence: 80%," it doesn't really help much.

NeuroScript fixes that. It uses Generative AI to turn tech jargon into easy-to-understand descriptions and tips that actually mean something to regular folks.

B. Architecture of the Generative AI Module

The Generative AI gets info from the prediction engine and XAI modules. It then passes this to the language model: the Predicted Class (ADHD, OCD, or Normal), Confidence Score, SHAP Feature Contributions, LIME Explanations, and Important Handwriting Characteristics. Using all that, the language model creates a structured diagnostic summary.

C. Personalized Recommendation Generation

The recommendation engine produces human-readable outputs including:

Diagnostic Summary

A concise explanation of the detected neurological pattern.

Behavioral Observations

Important handwriting characteristics identified during analysis.

Personalized Recommendations

Suggested actions such as:

- Clinical consultation
- Behavioral assessment
- Psychological screening
- Professional neurological evaluation

D. User-Friendly Explanation

Complex machine learning outputs are translated into simpler language for parents, teachers, healthcare pros, and general folks to grasp. An example from NeuroScript:

"The writing shows traits linked to ADHD, like uneven spaces and a bumpy writing flow. But while these hints point to possible focus issues, a full check-up is needed before deciding anything for sure. It's best to talk to a healthcare expert."

This makes things easier to understand and helps people get more out of the info.

E. Advantages of Generative AI Integration

The integration of Generative AI provides:

- Personalized feedback
- Easy-to-understand reports
- Improved user experience
- Enhanced communication between AI and end-users
- Greater accessibility for non-technical audiences

The Generative AI component represents a significant advancement beyond traditional classification systems by providing meaningful interpretation of prediction results.

6.8. REAL-TIME STREAMLIT DASHBOARD

A. System Interface

A user-friendly web application was developed using Streamlit to provide real-time access to the NeuroScript framework.

The dashboard serves as the primary interaction platform between users and the AI system.

B. Dashboard Features

The interface includes:

Handwriting Upload Module

Users can upload handwriting images directly through the dashboard.

Prediction Engine

Uploaded images are automatically processed and analyzed by the trained SVM model.

XAI Visualization Panel

SHAP and LIME explanations are displayed visually to explain prediction outcomes.

AI Recommendation Section

The Generative AI module produces personalized diagnostic summaries and recommendations.

Model Performance Dashboard

Users can view model evaluation statistics and comparison results.

C. Prediction Workflow

The complete prediction pipeline is:

Handwriting Upload

→ Image Preprocessing

→ Feature Extraction

→ SVM Classification

→ ADHD/OCD/Normal Prediction

→ SHAP Analysis

→ LIME Analysis

→ AI Recommendation Generation

→ Dashboard Output

D. Practical Significance

The Streamlit dashboard transforms NeuroScript into an accessible and deployable solution that can be used by:

- Researchers
- Healthcare professionals
- Educational institutions
- Parents
- Students

The system provides fast, transparent, and user-friendly neurological screening through handwriting analysis.

7. EXPERIMENTAL SETUP AND RESULTS DISCUSSION

7.1. Experimental Setup

We did all the NeuroScript framework experiments with Python's machine learning and deep learning tools. We used Jupyter Notebook, Visual Studio Code, and Streamlit for our work, so that's what the setup looked like.

The following technologies were used:

- Python Programming Language
- Scikit-learn
- TensorFlow/Keras
- NumPy
- Pandas
- OpenCV
- Matplotlib
- SHAP
- LIME
- Streamlit

The dataset was divided into training and testing sets using an 80:20 ratio. Three classification models, namely Random Forest, Support Vector Machine, and Convolutional Neural Network, were trained and evaluated using the same dataset to ensure fair comparison.

7.2. Performance Evaluation

The classification performance of the developed models was assessed using Accuracy, Precision, Recall, and F1-Score.

The experimental results indicate that Support Vector Machine achieved the highest overall performance among all evaluated classifiers.

TABLE III. PERFORMANCE COMPARISON OF CLASSIFIERS

Model	Accuracy	Precision	Recall	F1 Score
Random Forest	70%	74%	70%	70%
SVM	80%	80%	80%	80%
CNN	40%	35%	40%	33.33%

The results demonstrate that SVM outperformed both Random Forest and CNN across all evaluation metrics.

7.3. Comparative Analysis

The superior performance of SVM can be attributed to its strong capability of handling high-dimensional feature spaces and small datasets. Since the NeuroScript dataset contains only 48 handwriting samples, SVM was able to generalize more effectively than the CNN model.

Random Forest also produced satisfactory results due to its ensemble learning mechanism, which combines multiple decision trees to improve prediction stability. However, its accuracy remained lower than that of SVM.

CNN achieved the lowest performance because deep learning models generally require larger datasets for effective feature learning. The limited dataset size restricted the CNN model's ability to learn meaningful handwriting representations.

7.4. Discussion of Findings

The obtained results indicate that handwriting contains meaningful behavioral patterns associated with neurological conditions such as ADHD and OCD.

Several handwriting characteristics repeatedly influenced model predictions:

- Irregular spacing
- Inconsistent letter formation
- Variable writing speed
- Stroke inconsistency
- Pen movement behavior
- Alignment irregularities

These observations support previous findings in neurological handwriting research and demonstrate the feasibility of using handwriting as a non-invasive screening tool.

The strong performance of SVM suggests that machine learning-based handwriting analysis can provide valuable assistance in preliminary neurological assessment.

7.5. Practical Implications

The proposed NeuroScript framework offers several practical advantages:

Early Screening

The system can assist in the early identification of ADHD and OCD-related handwriting patterns.

Non-Invasive Assessment

Unlike many traditional neurological assessment methods, handwriting analysis does not require expensive equipment or invasive procedures.

Explainable Predictions

The integration of SHAP and LIME ensures transparency and helps users understand prediction outcomes.

Accessible Deployment

The Streamlit dashboard enables real-time usage through a simple web interface.

These characteristics make NeuroScript suitable for educational institutions, research environments, and preliminary healthcare screening applications.

8. LIMITATIONS

Though NeuroScript looks pretty good for spotting neurological issues through handwriting, we need to keep some downsides in mind. First off, the data they used is tiny – just 48 handwriting samples. That can limit how well the system works on other people's writing and raise the chance of it learning too closely from what it was given. Also, there aren't many 'Normal' samples; there are way more of those with ADHD and OCD. This skew might mess with how well it can correctly identify normal versus non-normal cases. Plus, all the samples came from a pretty narrow set of people, who might not reflect how everyone writes differently based on their age, schooling, or where they come from. Then, while NeuroScript can give writing-based advice using AI, it can't replace an official checkup with an actual doctor. Lastly, it solely relies on handwriting for its analyses – throwing in factors like behavior and physical symptoms could really boost its precision. To make this better, researchers ought to get more diverse data, look at various other neurological clues, make their models stronger, and see if their program stands up in bigger tests with real patients.

9. CONCLUSION

This research introduced NeuroScript, an Explainable AI-based handwriting analysis system for spotting ADHD and OCD early on. NeuroScript combines machine learning, deep learning, and uses XAI and Generative AI, along with a Streamlit dashboard, to make neurological screenings easier to understand and access. It looks at how fast you write, how consistent your pen strokes are, spacing patterns, letter formation, and overall pen movement. The study tested three models – Random Forest, Support Vector Machine, and Convolutional Neural Network. Results showed that the Support Vector Machine did best, hitting 80% accuracy, precision, recall, and F1-score. To boost transparency and trust, SHAP and LIME methods were plugged into the predictions. Also, a Generative AI part explains complex data in simpler diagnostic notes and

personal tips. The Streamlit dashboard can analyze handwriting right away and makes interacting with the system a breeze. So, NeuroScript shows how AI can help in simple, clear, and non-invasive ways for early neurological issues. Future efforts will expand the dataset and include bigger, more diverse groups. They'll also boost classification accuracy using better deep learning methods and do clinical tests to check how well it works in the real world.

10. ACKNOWLEDGEMENTS

We'd like to say a big thank you to Dr. Umara Zahid for the ongoing guidance, helpful feedback, and support through our research. Plus, we appreciate the Software Engineering Department at International Islamic University Islamabad for giving us the resources and great academic setting we needed for our project. Finally, a special thanks to everyone who gave handwriting samples and helped with data collection – your cooperation was super important to making this work come together.

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